

Assignment 1

February 3, 2026

Question 1. (30 points) Please discuss the following statements and briefly explain your opinion on them in a clear and concise way. Keep in mind that these are not true or false statements.

1. If a linear programming problem has a feasible solution, it is guaranteed to have an optimal solution.
2. If the feasible region of a linear programming problem is unbounded, it means the objective function can be improved indefinitely.
3. If an LP is feasible, then it has an optimal solution that is a corner point.
4. The graphical solution method can only be used to solve linear programs with two decision variables.

Question 2. (15 points) I also would like you to discuss these statements with your classmates and tell me how their opinion differ from yours (if they do). List out the uni's of every student you have discussed the statements with.

Question 3. (55 points) A shelter dedicated to supporting victims of human trafficking is tasked with procuring essential resources to provide comprehensive care. These resources include both cleaning supplies and personal care items, crucial for maintaining hygiene and health. The shelter procures these items from different suppliers, given in Table 1. Each month, the shelter places one order with each supplier.

Given the limited funding and different costs of resources from each supplier, the shelter must strategically plan its purchases to minimize expenses while ensuring it meets the necessary service standards for effective victim support. The minimum required quantities for these standards are listed in Table 3. The shelter has a \$2,000 monthly budget for all the resources.

Deliverables:

1. Include a clear description of the decision variables, the objective function, and each constraint. (10 points)
2. Provide your algebraic LP formulation. (20 points)
3. Solve the LP using any solver (e.g., Gurobi, CPLEX) and share the code/model you used for solving the problem. A screenshot would also work. (20 points)
4. Explain your solution. Discuss the implications of your solution regarding the shelter's operations and resource allocation strategies. (5 points)

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Item	Cleaning				Personal Care			
Supplier	Toilet Paper (1 roll)	Liquid Soap (lt.)	Detergent (lt.)	Cloth (m ²)	Tooth paste	Tooth brush	Pads	Shampoo (lt.)
1	\$0.80	\$6.40	\$6.80	\$10.00	\$2.60	\$0.80	\$0.20	\$2.30
2	\$0.95	\$3.98	\$4.60	\$11.00	\$3.00	\$0.85	\$0.18	\$1.20
3	\$0.84	\$5.50	\$7.50	\$10.50	\$2.80	\$0.82	\$0.15	\$3.00

Table 1: Cost of acquiring needed resources from different suppliers.

Item	Cleaning				Personal Care			
Supplier	Toilet Paper (rolls)	Liquid Soap (lt.)	Detergent (lt.)	Cloth (m ²)	Tooth paste	Tooth brush	Pads	Shampoo (lt.)
1	150	25	20	10	50	50	150	20
2	100	15	10	10	60	60	100	20
3	70	30	15	15	30	30	100	30

Table 2: Maximum supply capacity of cleaning and personal care items from different suppliers.

Type of Resource	Minimum Required Quantity (units)
Toilet Paper (rolls)	200
Liquid Soap (liters)	40
Detergent (liters)	30
Cloths (square meters)	20
Toothpaste (tubes)	100
Toothbrushes	100
Sanitary Pads (packs)	300
Shampoo (liters)	40

Table 3: Minimum required quantities of cleaning and personal care items for a shelter housing 20 people. Quantities are estimated for one month of supply.

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To solve the problem using a solver, you can use the dataset given below.

```
# Data: Supplier information, cost per unit (c_jk), and maximum supply
      capacity (s_jk)
suppliers = ['Supplier1', 'Supplier2', 'Supplier3']
resources = ['Toilet_Paper', 'Liquid_Soap', 'Detergent', 'Cloths',
            'Toothpaste', 'Toothbrushes', 'Sanitary_Pads', 'Shampoo']

# Cost and maximum supply capacity for each supplier and resource
cost_per_unit = {
    ('Supplier1', 'Toilet_Paper'): (0.80, 150),
    ('Supplier1', 'Liquid_Soap'): (6.40, 25),
    ('Supplier1', 'Detergent'): (6.80, 20),
    ('Supplier1', 'Cloths'): (10.00, 10),
    ('Supplier1', 'Toothpaste'): (2.60, 50),
    ('Supplier1', 'Toothbrushes'): (0.80, 50),
    ('Supplier1', 'Sanitary_Pads'): (0.20, 150),
    ('Supplier1', 'Shampoo'): (2.30, 20),
    ('Supplier2', 'Toilet_Paper'): (0.95, 100),
    ('Supplier2', 'Liquid_Soap'): (3.98, 15),
    ('Supplier2', 'Detergent'): (4.60, 10),
    ('Supplier2', 'Cloths'): (11.00, 10),
    ('Supplier2', 'Toothpaste'): (3.00, 60),
    ('Supplier2', 'Toothbrushes'): (0.85, 60),
    ('Supplier2', 'Sanitary_Pads'): (0.18, 100),
    ('Supplier2', 'Shampoo'): (1.20, 20),
    ('Supplier3', 'Toilet_Paper'): (0.84, 70),
    ('Supplier3', 'Liquid_Soap'): (5.50, 30),
    ('Supplier3', 'Detergent'): (7.50, 15),
    ('Supplier3', 'Cloths'): (10.50, 15),
    ('Supplier3', 'Toothpaste'): (2.80, 30),
    ('Supplier3', 'Toothbrushes'): (0.82, 30),
    ('Supplier3', 'Sanitary_Pads'): (0.15, 100),
    ('Supplier3', 'Shampoo'): (3.00, 30)
}

# Minimum required quantities as a dataset for a shelter housing 20 people
  for one month
minimum_quantities = {
    'Toilet_Paper': 200,
    'Liquid_Soap': 40,
    'Detergent': 30,
    'Cloths': 20,
    'Toothpaste': 100,
    'Toothbrushes': 100,
    'Sanitary_Pads': 300,
    'Shampoo': 40
}
```